

BAD702: SMD MODULE 2

Question 1:

Explain random sampling and selection bias. How do they affect data analysis? Give examples.

Answer

1. Introduction

In statistics and data analysis, a **sample** is a subset of data taken from a larger **population**. Since conclusions about a population are often made using a sample, the **method of selecting the sample** is extremely important.

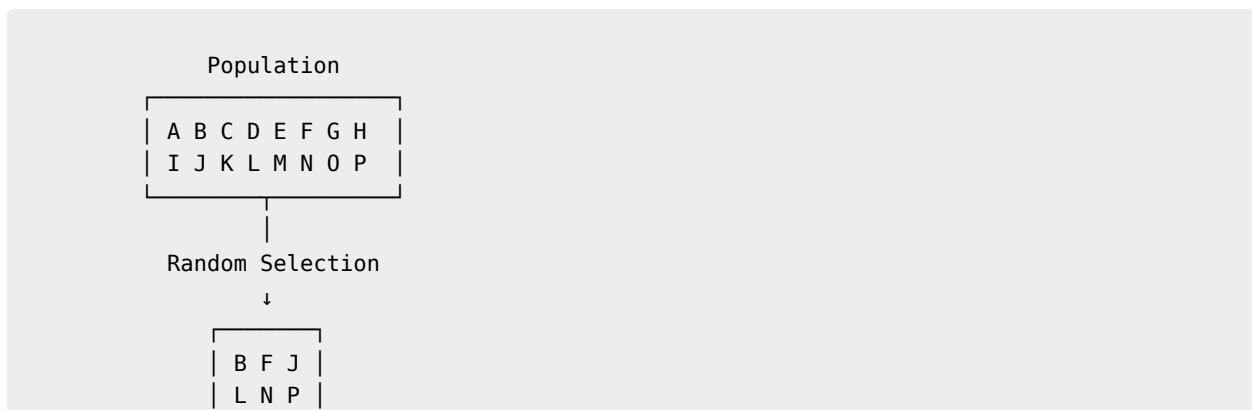
Random sampling helps obtain a representative sample and reduce bias, whereas **selection bias** occurs when the selected sample systematically differs from the population it is intended to represent.

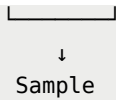
2. Random Sampling

Random sampling is a process in which **each available member of the population has an equal chance of being selected** for the sample at each draw.

The resulting sample is called a **simple random sample**.

Diagram





Random sampling reduces the possibility that the sample is systematically different from the population.

Types

1. Simple random sampling

- Sampling is performed randomly **without dividing the population into groups**.
- Every available member has an equal chance of selection.

2. Stratified sampling

- The population is divided into **strata**, which are homogeneous subgroups having common characteristics.
- Random samples are then taken from each stratum.

Example

Suppose a college has students from different years and we want a sample representing the college.

Instead of randomly selecting students from the entire population and potentially obtaining too few students from some years, the population can be divided into **year-wise strata** and students can be randomly selected from each stratum.

3. Selection Bias / Sample Bias

Bias is a **systematic error**.

Sample bias occurs when a sample **misrepresents the population** in a meaningful and nonrandom way.

The important point is that a random sample will not necessarily be exactly representative of the population because of random variation. It becomes **sample bias** when the difference is systematic and can be expected to continue for samples selected in the same way.

4. Classic Example: Literary Digest Poll

The famous **1936 US presidential election poll** demonstrates the importance of sampling method.

- **Literary Digest** polled more than **10 million people**.
- It predicted a victory for **Alf Landon**.
- **George Gallup** used scientifically selected samples of about **2,000 people** and correctly predicted **Franklin Roosevelt's** victory.
- The difference was not the size of the sample but **how the people were selected**.

The Literary Digest's sample included many people with relatively **high socioeconomic status**, including its subscribers and people appearing on lists associated with ownership of telephones and automobiles. Therefore, the sample did not properly represent the broader voting population.

Lesson

Large sample size does not guarantee an unbiased result

The **quality and representativeness of the sample** can be more important than its size.

5. Self-Selection Bias

A common form of selection bias is **self-selection bias**.

It occurs when people **choose themselves to participate** rather than being randomly selected.

Example

Reviews on websites such as Yelp are subject to self-selection bias because people voluntarily decide whether to submit reviews.

People motivated to write a review may:

- Have had a particularly **good or bad experience**.

- Have an association with the establishment.
- Differ systematically from people who do not submit reviews.

Therefore, the reviews may not accurately represent the experience of the entire customer population.

6. Effect on Data Analysis

Selection bias can seriously affect statistical and machine-learning analysis.

A. Incorrect Estimates

If the sample is not representative, calculated statistics such as the **sample mean** or **proportion** may not accurately estimate the corresponding population values.

B. Incorrect Conclusions

A biased sample can lead researchers to conclude that a pattern exists in the population when it does not, or miss a pattern that actually exists.

C. Biased Models

Statistical or machine-learning models trained on biased data can learn patterns that are not representative of the target population.

D. Poor Generalization

A model developed from a biased sample may perform well on that sample but perform poorly when applied to the broader population.

Therefore:

Biased sample → Biased analysis → Unreliable conclusions

7. Random Sampling vs Sample Bias

Feature	Random Sampling	Sample Bias
Meaning	Randomly selects members from population	Sample systematically misrepresents population

Feature	Random Sampling	Sample Bias
Selection	Based on random procedure	Selection favors certain groups
Representativeness	Helps improve representativeness	Poor representativeness
Effect	Reduces systematic bias	Produces systematic error
Example	Randomly selecting students	Polling only people with particular characteristics

8. Importance of Random Sampling

Random sampling remains important even in the era of **big data**.

It can:

- **Reduce bias.**
- Produce a more representative sample.
- Reduce the time and cost of data collection.
- Allow greater attention to **data exploration and data quality**.
- Make manual inspection of **missing data and outliers** more feasible.

The notes emphasize that **data quality can matter more than data quantity** when making an estimate or building a model.

Question 2:

Describe the concept of confidence intervals. How are they constructed using the normal distribution?

Answer

1. Introduction

A **confidence interval** is a statistical method used to express an estimate not as a single number, but as a **range of values** that accounts for **sampling uncertainty**.

A confidence interval is associated with a **confidence level**, such as **90%, 95% or 99%**. The confidence level represents the percentage of intervals, constructed in the same way from the same population, that are expected to contain the **statistic of interest**.

2. Need for Confidence Intervals

A sample statistic, such as the **sample mean**, varies from one sample to another. Therefore, a single point estimate does not indicate how uncertain the estimate is.

A confidence interval provides:

- A **point estimate**.
- A **lower interval endpoint**.
- An **upper interval endpoint**.
- An indication of the **uncertainty or sampling variability** of the estimate.

Thus,

$$\text{Confidence Interval} = \text{Estimate} \pm \text{Margin of Error}$$

3. Role of Normal Distribution

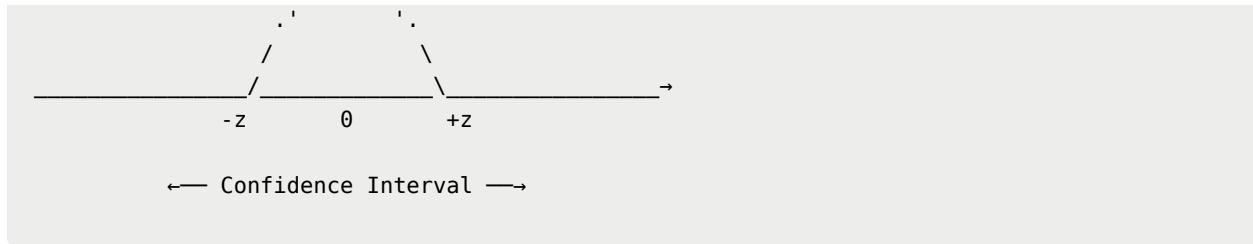
The **Central Limit Theorem (CLT)** states that the means obtained from repeated samples tend to have a **normal, bell-shaped sampling distribution** when the sample size is sufficiently large and the population is not too far from normal.

This makes the **normal distribution** useful for constructing confidence intervals.

Normal Distribution Diagram

Normal Distribution





For a normal distribution:

- About **68%** of observations lie within 1 standard deviation of the mean.
- About **95%** lie within 2 standard deviations.

4. Standard Error

The **standard error (SE)** measures the variability of a sample statistic over repeated samples.

For the sample mean:

$$SE = \frac{s}{\sqrt{n}}$$

where:

- s = sample standard deviation
- n = sample size.

As the **sample size increases**, the standard error decreases. Therefore, the estimate becomes more precise.  BAD702-module-2-textbook(1)

5. Construction of Confidence Interval Using Normal Distribution

For a normally distributed sampling statistic, the general form is:

$$\text{Confidence Interval} = \text{Sample Estimate} \pm \text{Critical Value} \times SE$$

For a sample mean, this can be written as:

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

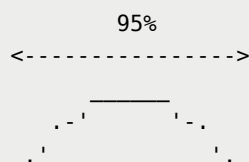
where:

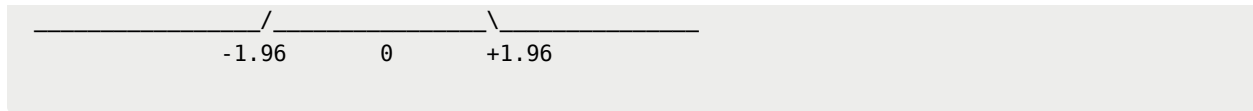
- $\bar{x}$ = sample mean
- s = sample standard deviation
- n = sample size
- $z_{\alpha/2}$ = critical value from the standard normal distribution
- $\alpha = 1 - \text{confidence level}$.

For example, for a **95% confidence level**:

$$\alpha = 1 - 0.95 = 0.05$$

and the central 95% of the normal distribution is used to determine the interval.





Hence, the interval can be expressed as:

$$\bar{x} \pm 1.96 \frac{s}{\sqrt{n}}$$

when the normal critical value is appropriate.

6. Steps for Construction

Step 1: Calculate the sample estimate

For example, calculate the sample mean:

$$\bar{x} = \frac{\sum x_i}{n}$$

Step 2: Calculate the standard error

$$SE = \frac{s}{\sqrt{n}}$$

Step 3: Select the confidence level

Common levels are:

- 90%
- 95%
- 99%

Step 4: Obtain the critical value

Use the appropriate value from the **standard normal distribution**.

Step 5: Calculate the margin of error

$$\text{Margin of Error} = z_{\alpha/2} \times SE$$

Step 6: Calculate the endpoints

$$\text{Lower limit} = \bar{x} - z_{\alpha/2}SE$$

$$\text{Upper limit} = \bar{x} + z_{\alpha/2}SE$$

Therefore:

$$CI = [\bar{x} - z_{\alpha/2}SE, \bar{x} + z_{\alpha/2}SE]$$

7. Interpretation

Suppose a sample gives:

$$\bar{x} = 50$$

and the calculated 95% confidence interval is:

$$[47, 53]$$

The interval gives a range of plausible values around the estimate:

$$[50 \pm 3]$$

The **confidence level** describes the long-run coverage of intervals constructed using the same procedure. It should not be interpreted simply as saying that there is a 95% probability that a fixed population parameter lies inside this particular interval.

8. Factors Affecting Confidence Interval Width

A. Confidence Level

A higher confidence level produces a wider interval.

Higher confidence $\Rightarrow$ Wider interval

For example, a 99% confidence interval is wider than a 95% interval.

B. Sample Size

A larger sample reduces standard error:

$$SE = \frac{s}{\sqrt{n}}$$

Therefore:

Larger sample $\Rightarrow$ Smaller SE $\Rightarrow$ Narrower interval

The notes describe this as the **square-root-of- n rule**: reducing standard error by a factor of 2 requires increasing the sample size by a factor of 4.  BAD702-module-2-textbook(1)

9. Important Note

The normal-distribution approach is based on the behavior of the **sampling distribution**. The **Central Limit Theorem** explains why sample means often become approximately normal for sufficiently large samples, even when the underlying population is not normally distributed.  BAD702-module-2-textbook(1)

The notes also emphasize that **bootstrap confidence intervals** are an effective alternative and do not require the same distributional assumptions.  BAD702-module-2-textbook(1)

Question 3:

Explain the Poisson distribution in detail and mention its applications.

Answer

1. Introduction

The **Poisson distribution** is a **discrete probability distribution** used to model the number of events occurring in a fixed **interval of time or space**, when the events occur randomly at a given overall rate. Examples include visitors arriving at a

website, cars arriving at a toll plaza, imperfections in fabric, and typos in a fixed number of lines of code.

2. Poisson Distribution

Let X denote the **number of events** occurring in a specified interval.

The Poisson distribution is characterized by a single parameter:

λ

where λ is the **average number of events per interval**.

The probability of observing exactly k events is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

where:

- X = number of events
- $k = 0, 1, 2, \dots$
- λ = average event rate
- $e \approx 2.71828$

The probabilities for all possible values of k sum to 1.

3. Conditions / Characteristics

The Poisson distribution is appropriate when:

1. We are counting the **number of events**.
2. Events occur randomly within a specified **time or spatial interval**.
3. The events have a given overall rate λ .
4. The rate λ is assumed to remain **constant over the period or region being considered**.
5. The occurrence of one event does not determine the occurrence of another event.

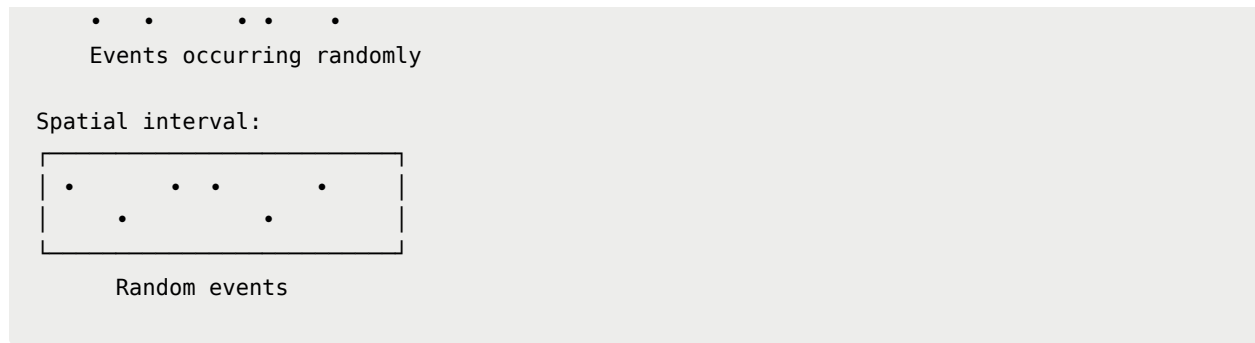
 BAD702-module-2-textbook(1)

Examples

Examples

Time interval:

0 $\longrightarrow$ T



4. Mean and Variance

For a Poisson distribution with parameter λ :

$$E(X) = \lambda$$

and

$$Var(X) = \lambda$$

Therefore:

$$\text{Mean} = \text{Variance} = \lambda$$

and the standard deviation is:

$$\sigma = \sqrt{\lambda}$$

This property is an important characteristic of the Poisson distribution.

5. Example

Suppose a website receives an average of **3 visitors per minute**.

Therefore:

$$\lambda = 3$$

The probability of exactly **2 visitors** arriving in one minute is:

$$\begin{aligned} P(X = 2) &= \frac{e^{-3} 3^2}{2!} \\ &= \frac{e^{-3} \times 9}{2} \end{aligned}$$

$$P(X = 2) \approx 0.224$$

Thus, there is approximately a **22.4% probability** of exactly two visitors arriving in one minute.

6. Applications of Poisson Distribution

The Poisson distribution is useful whenever we are interested in the **number of events occurring per unit of time or space**.

A. Website Traffic

It can model the number of:

- Visitors arriving at a website.
- Requests received by a server.
- Events occurring during a fixed time interval.

The notes specifically give **website visitors** as an example.

B. Traffic and Transportation

It can model the number of:

- Cars arriving at a **toll plaza**.
- Vehicles arriving at a particular location during a fixed time period.

C. Manufacturing

It can model the number of **imperfections or defects** in a fixed amount of material, such as imperfections in a square meter of fabric.

D. Software Engineering

It can model the number of **typos per 100 lines of code** or other count-based events occurring over a specified amount of code.

E. Engineering Reliability

Related event-count/rate models can be used in engineering applications involving **failures**. For rare events such as aircraft-engine failures, the event rate may be difficult to estimate because very little failure data may be available.

7. Important Assumption: Constant Rate

A key assumption in using the Poisson distribution is that the event rate

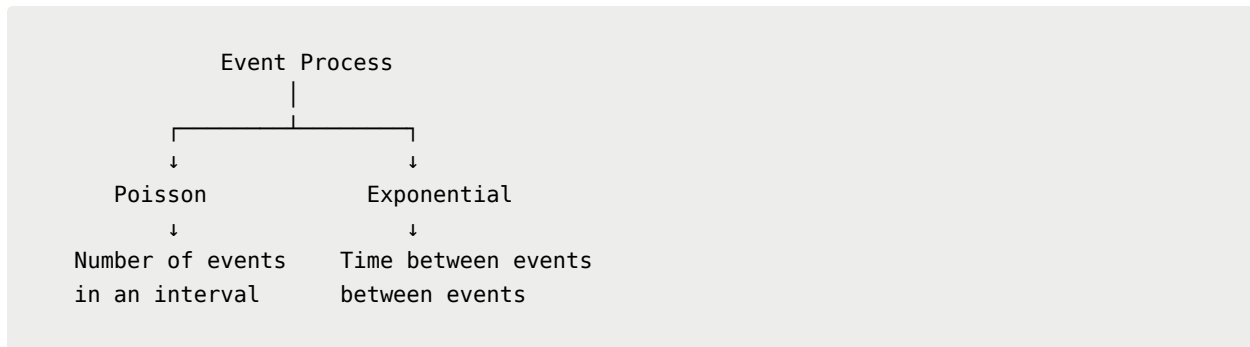
$$\lambda = \text{constant}$$

over the period or region being studied.

For example, traffic on roads or data-network traffic may vary by **time of day or day of week**. In such cases, one global Poisson rate may not be appropriate. The time period or spatial region can instead be divided into sufficiently homogeneous segments and modeled separately.

8. Poisson Distribution and Exponential Distribution

The **Poisson distribution** models the **number of events** occurring in a fixed interval, whereas the related **exponential distribution** models the **time between consecutive events**.



For example:

- **Poisson:** Number of cars arriving at a toll plaza in 10 minutes.
- **Exponential:** Time between successive cars arriving at the toll plaza.

9. Summary of Important Formulae

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

$$\sigma = \sqrt{\lambda}$$

where λ represents the average number of events per specified time or spatial interval.

Question 4:

Explain the normal, long-tailed, and Student's t-distributions in statistical modeling.

Answer

1. Introduction

Probability distributions are important in **statistical modeling** because they describe how data or statistical quantities are distributed. Three important distributions are the **normal distribution**, **long-tailed distributions**, and **Student's t-distribution**.

2. Normal Distribution

The **normal distribution** is a **bell-shaped, symmetric distribution** that has historically played an important role in statistics. It is particularly useful because many **sample statistics** have approximately normal sampling distributions.

Diagram



Characteristics

- It is **bell-shaped and symmetric**.
- The center represents the **mean**.
- Data is concentrated around the center.
- Approximately **68%** of the data lies within one standard deviation of the mean.
- Approximately **95%** lies within two standard deviations.

Standardization

A value can be converted to a **z-score** using:

$$z = \frac{x - \bar{x}}{s}$$

where:

- x = individual observation
- $\bar{x}$ = mean
- s = standard deviation.

The **standard normal distribution** has:

Mean = 0, Standard deviation = 1

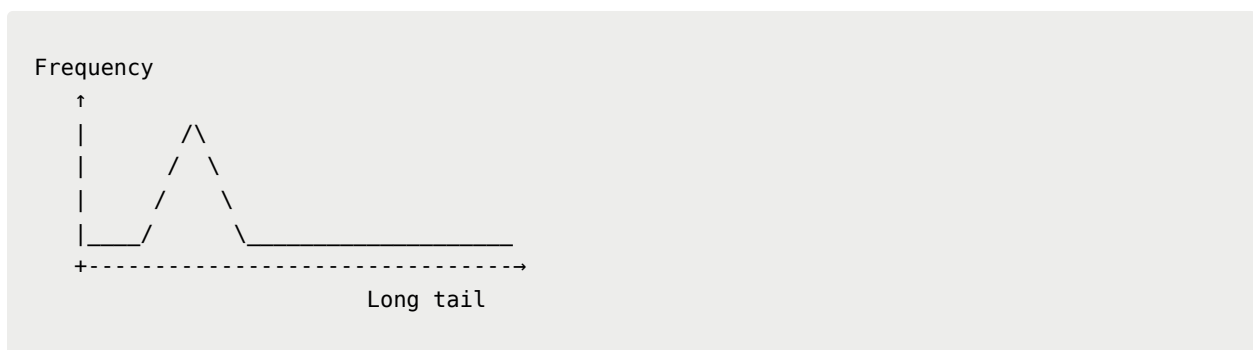
A **QQ-plot** can be used to determine how closely a sample follows a specified distribution such as the normal distribution. If the points approximately follow a diagonal line, the sample can be considered close to normal.

Importance in Modeling

The normal distribution is especially useful for modeling **errors and sample statistics**. However, most **raw data is not necessarily normally distributed**, so normality should not automatically be assumed.

3. Long-Tailed Distributions

A **long-tailed distribution** has a long, narrow portion or **tail** where relatively extreme values occur at low frequency.



Important Terms

Tail: The long narrow portion of a frequency distribution where relatively extreme values occur at low frequency.

Skew: A condition in which one tail of a distribution is longer than the other.

Characteristics

- Raw data is often **not normally distributed**.
- A distribution may be **highly skewed**, such as **income data**.
- Both symmetric and asymmetric distributions can have long tails.
- Tails represent **extreme small and large values**.
- Long tails are important because extreme events may occur more often than a normal distribution would predict.

Example

Stock returns are a good example of long-tailed data. The notes use **Netflix (NFLX) daily stock returns** to illustrate this behavior using a Q-plot.

Thus, assuming a normal distribution for data with long tails can underestimate the probability of extreme events.

4. Student's t-Distribution

The **Student's t-distribution** is a **family of distributions resembling the normal distribution but having thicker tails**.

Diagram



Characteristics

- It resembles the **normal distribution**.
 - It has **thicker tails** than the normal distribution.
 - It accounts for greater uncertainty associated with estimating a population parameter from a sample.
 - It is associated with **degrees of freedom**.
 - As the degrees of freedom increase, the t-distribution approaches the normal distribution.
-

5. Uses of Student's t-Distribution

The t-distribution is used as a reference distribution for:

- **Sample means**
- **Difference between two sample means**
- **Regression parameters**
- **Confidence intervals**
- Other statistical inference procedures.

For example, for a sample of size n , with sample mean $\bar{x}$ and standard deviation s , a 90% confidence interval can be expressed as:

$$\bar{x} \pm t_{n-1, 0.05} \frac{s}{\sqrt{n}}$$

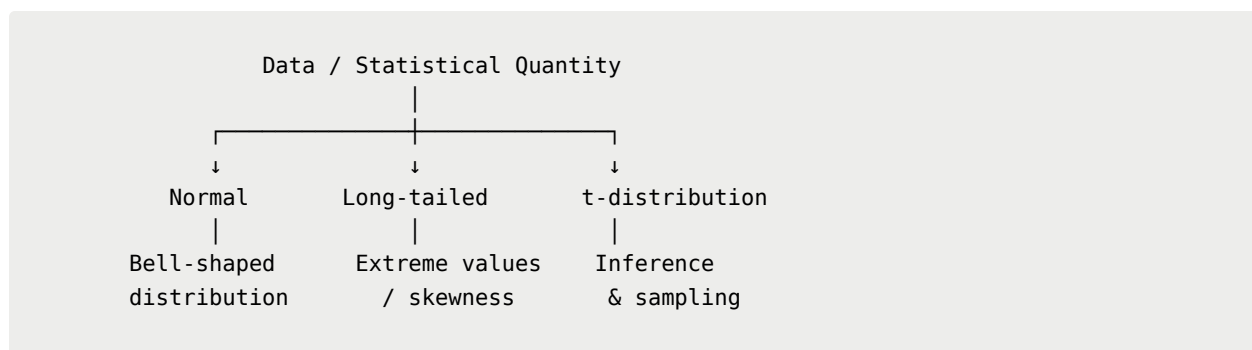
where $t_{n-1, 0.05}$ is the appropriate t-statistic with **$n-1$ degrees of freedom**.

6. Comparison

Feature	Normal Distribution	Long-Tailed Distribution	Student's t-Distribution
Shape	Bell-shaped and symmetric	Has long tails; may be symmetric or skewed	Bell-shaped, similar to normal
Tails	Relatively thin	Long/heavy tails	Thicker than normal
Main focus	Errors and sampling distributions	Extreme values in real data	Statistical inference
Example	Sample statistics	Income, stock returns	Sample mean, regression parameters
Main use	Modeling and approximation	Modeling extreme observations	Confidence intervals and inference

7. Importance in Statistical Modeling

These distributions should be selected based on the **nature of the data and the modeling objective**.



The normal distribution is powerful for many **sampling distributions**, while long-tailed distributions are important for understanding **extreme observations** in real-world data. The t-distribution provides a useful reference for **sampling variability and statistical inference**, especially for sample means and regression parameters.

Question 5:

Explain the concept of a sampling distribution. Discuss how a sampling distribution is formed and explain its importance in statistical inference.

Answer

1. Definition

A **sampling distribution** is the **frequency distribution of a sample statistic over many samples or resamples drawn from the same population**.

For example, if we repeatedly draw samples from a population and calculate the **sample mean** for each sample, the distribution of all these sample means is called the **sampling distribution of the sample mean**.

It is important to distinguish:

- **Data distribution** → distribution of individual data values.
 - **Sampling distribution** → distribution of a **sample statistic** over many samples.
-

2. Formation of a Sampling Distribution

A sampling distribution can be formed as follows:

Step 1: Select a population

Consider a population from which samples are to be drawn.

Step 2: Draw a sample

Take a sample of size n from the population.

Step 3: Calculate the statistic

Calculate a statistic such as:

$$\bar{x} = \frac{\sum x_i}{n}$$

for the sample.

Step 4: Repeat sampling

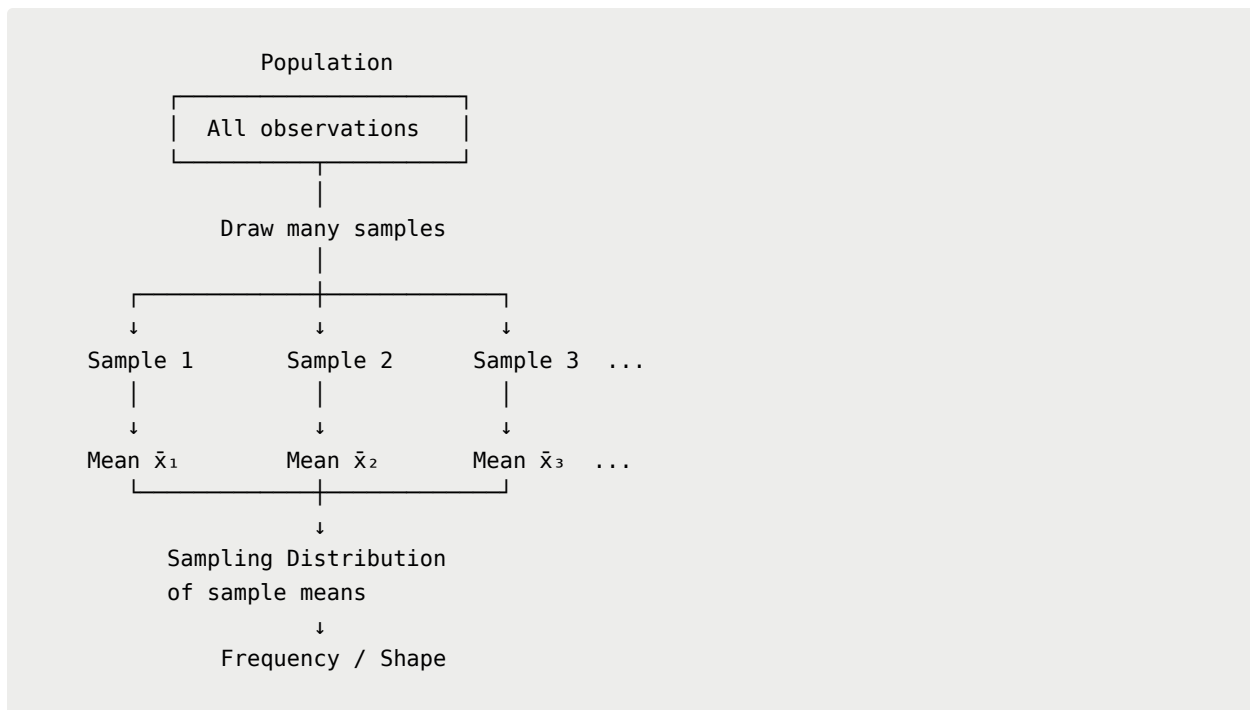
Draw many different samples of the same size and calculate the same statistic for every sample.

Step 5: Form the distribution

Plot the resulting statistics using a **frequency distribution or histogram**.

The resulting distribution is the **sampling distribution**.

Diagram



The key idea is that the **sample statistic changes from sample to sample**, producing sampling variability.

3. Sampling Variability

Because different samples contain different observations, the value of a sample statistic will generally differ from one sample to another.

For example:

$$\bar{x}_1=52, \bar{x}_2=49, \bar{x}_3=51, \bar{x}_4=54$$

These different values form a distribution.

Therefore, the sampling distribution tells us **how a statistic would turn out differently from sample to sample**.

4. Standard Error

The **standard error (SE)** measures the variability of a sample statistic over many samples.

For the sample mean:

$$SE = \frac{s}{\sqrt{n}}$$

where:

- s = sample standard deviation
- n = sample size.

The standard error should not be confused with standard deviation:

Standard deviation → variability of individual data values

Standard error → variability of a sample statistic

As the sample size increases:

$$n \uparrow \Rightarrow SE \downarrow$$

Thus, larger samples generally produce more precise estimates. The notes describe this as the **square-root-of- n rule**: to reduce the standard error by a factor of 2, the sample size must be increased by a factor of 4.

 BAD702-module-2-textbook(1)

5. Sampling Distribution and Central Limit Theorem

The **Central Limit Theorem (CLT)** states that the sampling distribution of sample means tends to take on a **normal, bell-shaped form as sample size increases**, provided the underlying population is not excessively far from normal.

```
Small / repeated samples
    ↓
Sample statistics
    ↓
Distribution of statistics
    ↓
As sample size increases
    ↓
Approximately normal shape
```

This is important because many statistical formulas are based on the approximate normality of sampling distributions.

6. Importance in Statistical Inference

Sampling distributions are fundamental to **statistical inference**, because we usually have only a sample but want to make conclusions about a much larger population.

A. Estimating Sampling Error

The sampling distribution shows how much a statistic can vary between different samples.

Its standard deviation, called the **standard error**, quantifies this variability.

B. Confidence Intervals

Sampling distributions provide the basis for constructing **confidence intervals**.

For example, using an appropriate reference distribution, an interval can be constructed around a sample mean:

$$\bar{x} \pm \text{critical value} \times \text{SE}$$

Thus, sampling distributions allow us to quantify the uncertainty associated with a sample estimate.

C. Hypothesis Testing

Sampling distributions are also used to determine how unusual an observed statistic would be under a specified hypothesis.

Therefore, they provide the mathematical foundation for **hypothesis tests**. The notes specifically identify confidence intervals and hypothesis tests as major applications of sampling distributions and the Central Limit Theorem.

7. Bootstrap Estimation of Sampling Distribution

In practice, drawing many brand-new samples from the population is usually **not feasible**.

The **bootstrap** provides an alternative:

1. Draw a sample of size n **with replacement** from the available sample.
2. Calculate the statistic.
3. Repeat this process many times.
4. Examine the resulting distribution of statistics.

This gives an estimate of the **sampling distribution** without requiring new samples from the population.

```
Original Sample
    ↓
Resample with replacement
    ↓
Calculate statistic
    ↓
Repeat many times
    ↓
Bootstrap Sampling Distribution
```

The bootstrap can be used for virtually any statistic and does not depend on the Central Limit Theorem or normality assumptions.

8. Summary

Concept	Meaning
Population	Entire group of interest
Sample	Subset drawn from population
Sample statistic	Metric calculated from a sample
Sampling distribution	Distribution of a statistic over many samples
Sampling variability	Variation in the statistic from sample to sample
Standard error	Standard deviation of the sampling distribution
CLT	Sampling distribution tends toward normality as sample size increases
Bootstrap	Estimates sampling distribution using resampling with replacement

Question 6:

Explain the Central Limit Theorem and discuss its significance in statistical inference and the construction of confidence intervals and hypothesis tests.

Answer

1. Definition

The **Central Limit Theorem (CLT)** states that the **means drawn from multiple samples tend to resemble a bell-shaped normal distribution**, even when the source population itself is not normally distributed, provided that the **sample size is sufficiently large** and the departure from normality is not too great.

Thus, the CLT explains why the **sampling distribution of the sample mean** is often approximately normal.

2. Concept of CLT

Consider a population from which many samples of the same size are selected.

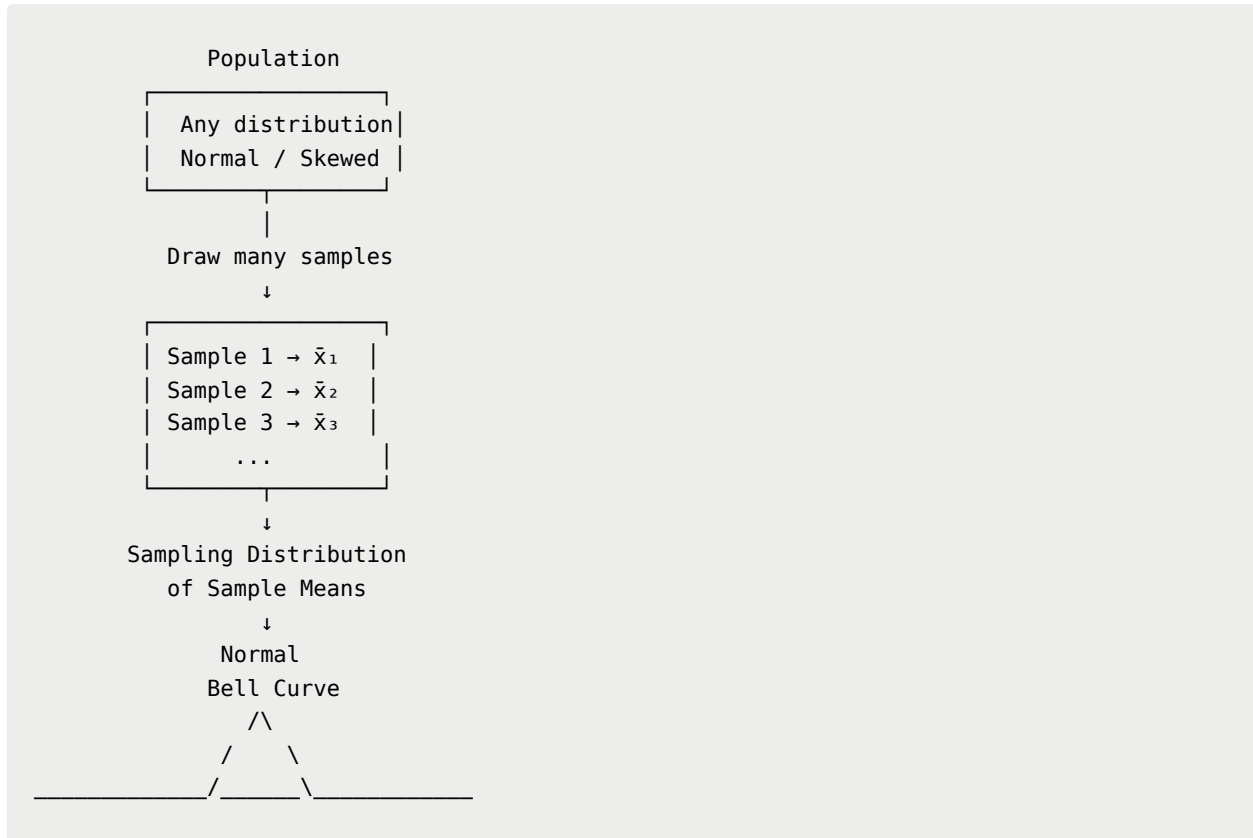
For each sample, calculate its mean:

$$\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots$$

These sample means form a **sampling distribution**.

As the sample size increases, this distribution tends to become approximately **normal (bell-shaped)**.

Diagram



The key point is:

Non-normal population → approximately normal sampling distribution
when the sample size is sufficiently large.

3. Standard Error and CLT

The **standard error** measures the variability of a sample statistic over repeated samples.

For the sample mean:

$$SE = \frac{s}{\sqrt{n}}$$

where:

- s = sample standard deviation
- n = sample size.

As n increases:

$$n \uparrow \Rightarrow SE \downarrow$$

Therefore, larger samples produce a sampling distribution with **less variability** and hence more precise estimates. BAD702-module-2-textbook(1)

This relationship is sometimes called the **square-root-of- n rule**. To reduce the standard error by a factor of 2, the sample size must be increased by a factor of 4. BAD702-module-2-textbook(1)

4. Significance in Statistical Inference

The main importance of the CLT is that it provides a theoretical basis for making **inferences about a population from a sample**.

We usually cannot collect the entire population. Instead:

Population → Sample → Sample statistic → Inference about population

Because the sampling distribution of many statistics can be approximated using the normal distribution, we can quantify **sampling variability and uncertainty**.

The CLT therefore supports the use of formulas for **sampling distributions**, particularly those used in classical statistical inference.

5. Role in Confidence Intervals

A **confidence interval** gives an estimate together with a range representing sampling uncertainty.

The general form is:

$$\text{Estimate} \pm \text{Critical Value} \times SE$$

For a sample mean, an appropriate reference distribution can be used to construct an interval around $\bar{x}$.

For example, using the **t-distribution**:

$$\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

The CLT is important because it explains why the sampling distribution of the sample mean can often be treated as approximately normal, allowing normal-approximation formulas and related reference distributions to be used for inference.

Effect of sample size

Since:

$$SE = \frac{s}{\sqrt{n}}$$

a larger sample gives a smaller standard error and consequently a **narrower confidence interval**.

$$\text{Larger } n \Rightarrow \text{smaller SE} \Rightarrow \text{more precise interval}$$

6. Role in Hypothesis Testing

In **hypothesis testing**, we want to determine whether an observed sample result is consistent with a specified hypothesis.

The CLT allows the sampling distribution of a statistic to be approximated by a known distribution.

Basic procedure

1. State the **null hypothesis**.
2. Obtain the sample and calculate the required statistic.
3. Determine the **sampling variability/standard error**.
4. Use the appropriate reference distribution.
5. Determine how unusual the observed statistic is under the null hypothesis.
6. Make a conclusion regarding the hypothesis.

The CLT therefore provides the basis for the **normal-approximation formulas used in hypothesis tests**. The notes specifically state that the CLT allows such formulas to be used for **confidence intervals and hypothesis tests**.

7. Relation with Student's t-Distribution

The **t-distribution** is a family of distributions resembling the normal distribution but with **thicker tails**.

It is widely used as a reference distribution for:

- **Sample means**
- **Differences between two sample means**
- **Regression parameters**
- Other statistical procedures.

The connection with CLT is important because sample statistics are often approximately normally distributed even when the underlying population is not. This helped establish the use of the t-distribution in classical statistical inference.

8. Important Limitation

The CLT does **not** say that the original population becomes normal.

It says that the **sampling distribution of the statistic**, particularly the sample mean, tends toward a normal shape as the sample size increases.

Also, if the underlying population is **extremely non-normal**, a larger sample may be required before the approximation becomes reliable.

In modern statistics, the **bootstrap** can also be used to estimate sampling distributions and standard errors without relying on the CLT or normality assumptions.

9. Summary

Aspect	Role of CLT
Sampling distribution	Makes sample means approximately normal for sufficiently large samples
Standard error	Quantifies variability of the sample statistic
Confidence intervals	Supports normal-approximation methods for estimating uncertainty
Hypothesis tests	Provides a reference distribution for assessing observed statistics
Sample size	Larger n generally gives smaller standard error
Student's t-distribution	Common reference distribution for sample means and related inference

Question 7:

Explain the concept of standard error. Derive/discuss its relationship with sample size and explain the square-root-of-n rule.

Answer

1. Definition of Standard Error

Standard Error (SE) is a measure of the **variability of a sample statistic over repeated samples drawn from the same population**.

For example, if many samples are drawn from a population and the **sample mean** is calculated for each sample, the standard deviation of those sample means is called the **standard error of the mean**.

Thus:

Standard Error = Standard deviation of the sampling distribution

It measures the **precision of a sample estimate**. A smaller standard error indicates that the estimate varies less from sample to sample.

2. Standard Deviation vs Standard Error

These two quantities should not be confused.

Standard Deviation	Standard Error
Measures variability of individual observations	Measures variability of a sample statistic
Describes the spread of the data	Describes uncertainty in the estimate
Does not necessarily decrease when sample size increases	Generally decreases as sample size increases
Example: spread of individual heights	Example: variability of sample means

3. Standard Error of the Sample Mean

For a sample of size n , with sample standard deviation ss , the standard error of the sample mean is:

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}$$

where:

- $SE(\bar{x})$ = standard error of sample mean
- s = sample standard deviation
- n = sample size.

If the population standard deviation σ is known, the theoretical expression is:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

The sample version uses s as an estimate of σ . BAD702-module-2-textbook(1)

4. Relationship Between Standard Error and Sample Size

From:

$$SE = \frac{s}{\sqrt{n}}$$

assuming s remains approximately constant:

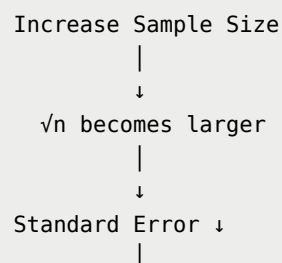
$$SE \propto \frac{1}{\sqrt{n}}$$

Therefore:

$$n \uparrow \Rightarrow SE \downarrow$$

As the sample size increases, the sample mean becomes **more precise** because its sampling variability decreases.

Diagram



↓
More precise estimate

5. Derivation of the Relationship

Suppose $X_1, X_2, \dots, X_n$ are independent observations with population variance σ^2 .

The sample mean is:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

The variance of the sample mean is:

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{X_1 + \dots + X_n}{n}\right)$$

For independent observations:

$$\text{Var}(\bar{X}) = \frac{1}{n^2} [\text{Var}(X_1) + \dots + \text{Var}(X_n)]$$

Since each observation has variance σ^2 :

$$\begin{aligned}\text{Var}(\bar{X}) &= \frac{1}{n^2} (n\sigma^2) \\ &= \frac{\sigma^2}{n}\end{aligned}$$

Taking the square root:

$$\boxed{SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}}$$

In practice, σ is generally unknown, so it is estimated using s :

$$\boxed{SE(\bar{x}) = \frac{s}{\sqrt{n}}}$$

6. Square-Root-of- n Rule

The **square-root-of- n rule** states that the standard error decreases in proportion to the **square root of the sample size**.

$$SE \propto \frac{1}{\sqrt{n}}$$

This has an important practical consequence:

To reduce the standard error by a factor of 2, the sample size must be increased by a factor of 4.

 BAD702-module-2-textbook(1)

Example

Suppose:

$$n = 100$$

Then:

$$\sqrt{100} = 10$$

If the sample size is increased to:

$$n = 400$$

then:

$$\sqrt{400} = 20$$

Therefore:

$$SE_{400} = \frac{s}{20} = \frac{1}{2} \frac{s}{10}$$

Hence:

$$SE_{400} = \frac{SE_{100}}{2}$$

So quadrupling the sample size halves the standard error.

7. Effect of Different Sample Sizes

Assume $s = 20$.

Sample Size n	Standard Error $SE = \frac{20}{\sqrt{n}}$ 
25	4
100	2
400	1
1600	0.5

This demonstrates that increasing the sample size improves precision, but with **diminishing returns**.

For example, reducing SE from 2 to 1 requires increasing the sample size from 100 to 400.

8. Importance in Statistical Inference

Standard error is important because it quantifies the **sampling uncertainty** of an estimate.

It is used in:

A. Confidence Intervals

A confidence interval has the general form:

$$\text{Estimate} \pm \text{Critical Value} \times SE$$

Therefore, a smaller SE generally produces a **narrower confidence interval**.

B. Hypothesis Testing

Standard error is used to standardize the difference between an observed estimate and a hypothesized value.

A general standardized statistic has the form:

$$\frac{\text{Observed estimate} - \text{Hypothesized value}}{SE}$$

This allows the observed result to be compared with an appropriate reference distribution.

9. Key Points

- **Standard error** measures the variability of a sample statistic over repeated samples.
- For the sample mean:

$$SE = \frac{s}{\sqrt{n}}$$

- Standard error decreases as **sample size increases**.
- The relationship is:

$$SE \propto \frac{1}{\sqrt{n}}$$

- **Doubling n** reduces SE by a factor of $\sqrt{2}$.
- **Quadrupling n** reduces SE by a factor of 2.
- Standard error is fundamental to **confidence intervals and hypothesis testing**.

Question 8:

Explain the bootstrap method in detail. Describe the procedure for generating bootstrap samples and discuss its applications in estimating the variability of a statistic.

Answer

1. Definition

The **bootstrap** is a resampling method used to estimate the **sampling distribution of a statistic** or model parameters.

In this method, additional samples are drawn **with replacement from the observed sample itself**, and the statistic is recalculated for each resample. It does not necessarily require the data or statistic to follow a normal distribution.

The bootstrap is particularly useful because collecting many brand-new samples from the population is generally **not feasible**.

2. Basic Idea of Bootstrap

Suppose the original observed sample is:

$$\mathbf{X} = \{x_1, x_2, x_3, \dots, x_n\}$$

The bootstrap treats this observed sample as a hypothetical population and repeatedly samples from it **with replacement**.



A **bootstrap sample** is therefore a sample taken **with replacement from an observed data set**.

3. Procedure for Generating Bootstrap Samples

For a sample of size n , the procedure for estimating the variability of a statistic is:

Step 1: Start with the observed sample

Consider:

$$X = \{x_1, x_2, \dots, x_n\}$$

Calculate the original statistic, such as the mean or median.

Step 2: Draw one observation

Randomly select one observation from the sample.

Step 3: Replace the observation

After recording it, **return it to the sample**. Therefore, the same observation can be selected again.

Step 4: Repeat n times

Continue sampling with replacement until a bootstrap sample of size n is obtained.

Step 5: Calculate the statistic

Calculate the statistic for this bootstrap sample.

For example:

$$T_1 = \text{mean of Bootstrap Sample 1}$$

Step 6: Repeat the entire process

Repeat Steps 2–5 **R times** to obtain:

T1,T2,...,TR

These R values form the **bootstrap distribution** of the statistic.

4. Example

Consider the observed sample:

$X=\{10,20,30,40,50\}$

Here:

$n=5$

One possible bootstrap sample is:

$\{20,20,40,10,50\}$

Notice that **20 occurs twice** while **30 is not selected**. This is possible because sampling is performed **with replacement**.

Its mean is:

$$T_1 = \frac{20 + 20 + 40 + 10 + 50}{5} = 28$$

Another bootstrap sample might be:

$$\{30, 50, 50, 10, 30\}$$

with mean:

$$T_2 = \frac{30 + 50 + 50 + 10 + 30}{5} = 34$$

Repeating this process many times gives:

$$T_1, T_2, T_3, \dots, T_R$$

The distribution of these values estimates the **sampling distribution of the mean**.

5. Estimating Standard Error

One major application of the bootstrap is estimating the **standard error** of a statistic.

After generating R bootstrap statistics:

$$T_1, T_2, \dots, T_R$$

calculate their standard deviation:

$$SE_{\text{bootstrap}} = SD(T_1, T_2, \dots, T_R)$$

This standard deviation is used as an estimate of the **standard error of the statistic**.

 BAD702-module-2-textbook(1)

Thus:

6. Number of Bootstrap Iterations

The number of repetitions R is the number of bootstrap iterations.

A larger R generally gives a **more accurate estimate** of the standard error or confidence interval.

 BAD702-module-2-textbook(1)

For example:

$$R = 1000$$

means that **1000 bootstrap samples** are generated and the statistic is calculated for each.

7. Applications of Bootstrap

A. Estimating Variability

The primary application is estimating the **sampling variability of a statistic**.

It can be applied to virtually any statistic without requiring a mathematical approximation to its sampling distribution.

Examples include:

- **Mean**
- **Median**
- Other sample statistics.

B. Estimating Standard Error

The standard deviation of the bootstrap statistics provides an estimate of the **standard error**.

This is useful when the standard error is difficult to obtain analytically.

 BAD702-module-2-textbook(1)

C. Confidence Intervals

Bootstrap distributions can be used to construct **confidence intervals**.

For an $x\%$ confidence interval:

1. Generate R bootstrap statistics.
2. Sort the bootstrap results.
3. Remove:

$$\frac{100 - x}{2} \%$$

from **each end**.

4. The remaining endpoints provide the bootstrap confidence interval.

 BAD702-module-2-textbook(1)

For example, for a **90% confidence interval**, remove 5% from each tail.

D. Model Parameters

Bootstrap resampling can also be applied to **multivariate data**, with rows sampled as units.

A model can then be fitted to each bootstrap sample to examine the **stability and variability of model parameters**.

 BAD702-module-2-textbook(1)

E. Predictive Modeling and Bagging

Bootstrap samples are also used with predictive models.

For example, multiple **classification and regression trees** can be trained on bootstrap samples. Their predictions can then be:

- **Averaged** for regression.
- Combined using **majority voting** for classification.

This technique is called **bagging** and can outperform a single model.

8. Advantages of Bootstrap

1. **Simple and effective** resampling method.
2. Estimates sampling distributions directly from the observed data.
3. Can be used for **virtually any statistic**.
4. Does not require the statistic to have a **normal distribution**.
5. Does not rely on the **Central Limit Theorem** or extensive mathematical approximations.
6. Can be used for estimating **standard errors and confidence intervals**.
7. Can also be applied to **model parameters and predictive models**.

9. Bootstrap vs Traditional Sampling

Aspect	Traditional Sampling	Bootstrap
Source of samples	Population	Observed sample
Sampling	New samples from population	Samples from observed data
Replacement	Depends on sampling design	With replacement
Main purpose	Obtain information about population	Estimate sampling distribution
Practicality	May require costly/new data	Computationally convenient
Distribution assumptions	May require theoretical assumptions	Often avoids distributional assumptions

Question 9:

Explain the different methods of constructing confidence intervals using resampling/bootstrap techniques. Discuss their interpretation.

Answer

1. Introduction

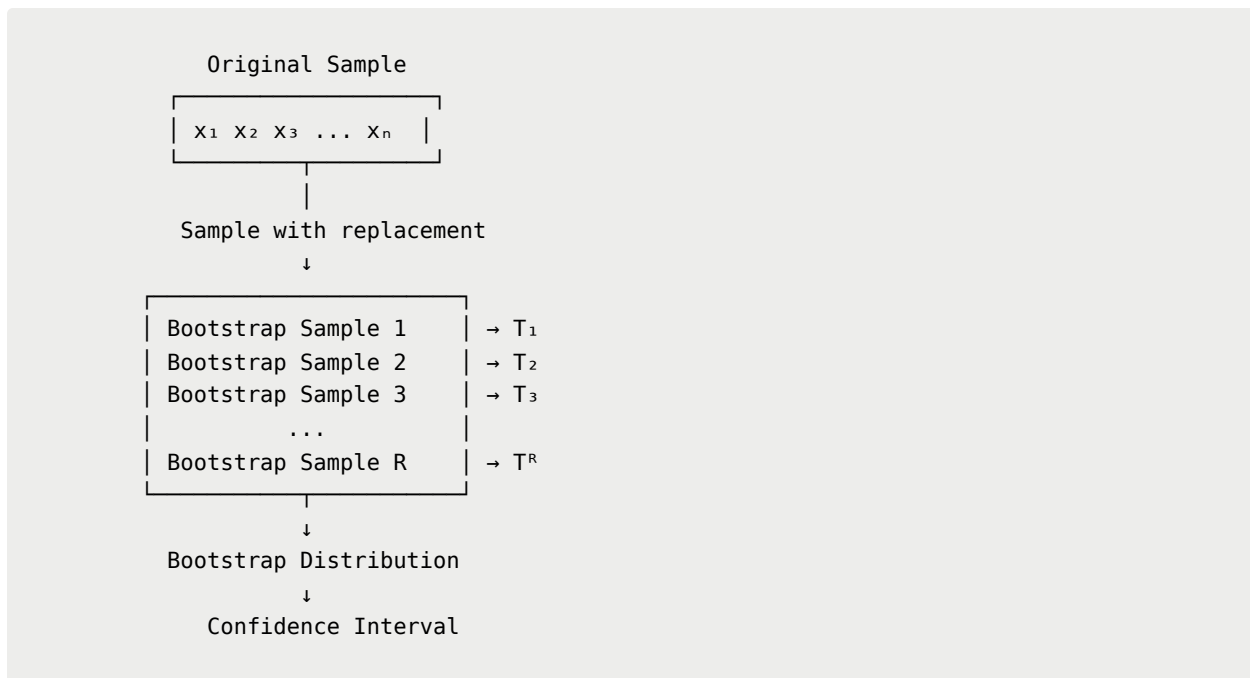
A **confidence interval** presents a sample estimate as a **range** rather than as a single number, thereby communicating the uncertainty associated with sampling variation.

The **bootstrap** is an effective method for constructing confidence intervals because it estimates the **sampling distribution of a statistic** by repeatedly resampling from the observed data. It does not necessarily require the data or statistic to be normally distributed.

2. Bootstrap Method for Confidence Intervals

Suppose the observed data contains n observations and the statistic of interest is T .

Diagram



The RR calculated statistics form the **bootstrap sampling distribution**. The confidence interval is then obtained from this distribution.

3. Percentile Bootstrap Confidence Interval

The **percentile method** is the method described in the notes for constructing a bootstrap confidence interval.

Procedure

Given a sample of size n :

1. Draw a random sample of size n **with replacement** from the observed data.
2. Calculate and record the statistic of interest.
3. Repeat Steps 1 and 2 many times, giving R bootstrap statistics.
4. Arrange the R statistics in increasing order.
5. For an $x\%$ confidence interval, remove

$$\frac{100 - x}{2} \%$$

from **each end** of the bootstrap distribution.

6. The two remaining **trim points** become the **interval endpoints**.

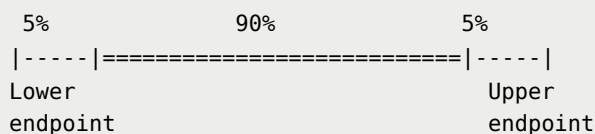
 BAD702-module-2-textbook(1)

Example: 90% Confidence Interval

For a 90% confidence interval:

$$\frac{100 - 90}{2} = 5\%$$

Therefore, remove the lowest 5% and highest 5% of bootstrap statistics.



Thus:

$$\text{CI } 90\% = [\text{5th percentile}, \text{95th percentile}]$$

4. Example

Suppose the statistic of interest is the **mean annual income** of a sample.

After generating many bootstrap samples, assume the bootstrap distribution of the sample mean is obtained.

For a **90% confidence interval**, the lowest 5% and highest 5% of the bootstrap means are removed.

If the two resulting endpoints are:

\$55,000 and \$70,000

then:

90% CI = [\$55,000, \$70,000]

The notes illustrate this procedure using a sample of loan applicants and their mean annual income.

5. Interpretation of a Bootstrap Confidence Interval

The **confidence level** indicates the long-run coverage of the interval-producing procedure.

For an x% confidence interval, when the **same sampling procedure is repeatedly followed**, the resulting intervals are expected, on average, to contain the statistic of interest approximately x% of the time.

For example, a **90% bootstrap confidence interval** means that the procedure is designed to capture the relevant value in approximately **90% of repeated samples**, in the long run.

It should **not** be interpreted simply as saying that there is a 90% probability that a fixed parameter is inside this particular calculated interval.

6. Why Bootstrap Confidence Intervals Are Useful

1. Does not require normality

The bootstrap does not necessarily require the statistic or underlying data to be normally distributed.

2. Applicable to many statistics

It can be used for **virtually any statistic**, making it flexible when mathematical formulas for sampling distributions are difficult.

3. Directly estimates sampling distribution

Instead of relying entirely on theoretical assumptions, the bootstrap **empirically estimates the sampling distribution** from the observed data.

4. Useful for uncertainty estimation

It provides a practical way to communicate the **uncertainty and variability of a sample estimate**.

7. Effect of Confidence Level

The width of the confidence interval depends on the selected confidence level.

Higher confidence level $\Rightarrow$ Wider interval

For example:

$$90\% CI < 95\% CI < 99\% CI$$

in terms of expected width, assuming the same data and method.

This is because a higher confidence level requires retaining a larger portion of the bootstrap distribution.

 BAD702-module-2-textbook(1)

8. Bootstrap Confidence Interval vs Formula-Based Interval

Feature	Formula-Based CI	Bootstrap CI
Basis	Mathematical/reference distribution	Empirical bootstrap distribution

Feature	Formula-Based CI	Bootstrap CI
Sampling	Uses theoretical sampling distribution	Resamples observed data
Distribution assumptions	May require assumptions	Does not necessarily require normality
Procedure	Formula using SE and critical value	Repeated resampling and percentile calculation
Flexibility	Limited for complicated statistics	Can be used for virtually any statistic
Computational requirement	Relatively low	Requires repeated computation

The notes emphasize that although classical methods use distributions such as the **t-distribution**, empirical bootstrap sampling can answer most questions about sampling error.

9. Important Points to Remember

Bootstrap CI = central portion of bootstrap distribution

For an $x\%$ confidence interval:

$$\text{Trim each tail by } \frac{100 - x}{2} \%$$

For example:

Confidence Level	Lower Tail	Upper Tail	
90%	5%	5%	
95%	2.5%	2.5%	
99%	0.5%	0.5%	

Question 10:

Explain the important probability distributions used in statistical modeling, including Binomial, Chi-square and F distributions, and discuss their

applications.

Answer

1. Introduction

Probability distributions describe the possible outcomes and their associated probabilities. They are important in statistical modeling because they provide a mathematical basis for **modeling data, sampling variability and hypothesis testing**.

Important distributions used in statistical modeling include the **Binomial, Chi-square and F-distributions**.

2. Binomial Distribution

The **Binomial distribution** is a discrete probability distribution that gives the distribution of the **number of successes in a fixed number of trials**, where each trial has two possible outcomes.

Key Terms

- **Trial:** An event with a discrete outcome.
- **Success:** The outcome of interest.
- **Binomial:** Having two outcomes, such as **yes/no** or **0/1**.
- **Binomial trial:** A trial with two outcomes.
- **Binomial distribution:** Distribution of the number of successes in n trials.

n

Probability Formula

If there are n trials and the probability of success in each trial is p , then the probability of exactly x successes is:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

where:

- n = number of trials
- x = number of successes
- p = probability of success
- $1 - p$ = probability of failure.

Example

Suppose the probability that a click converts to a sale is $p = 0.02$, and there are 200 clicks.

The binomial distribution can be used to calculate the probability of observing **0 sales in 200 clicks**.

 BAD702-module-2-textbook(1)

Applications

The Binomial distribution is useful for **binary outcomes**, such as:

- Buy / don't buy
- Fraud / no fraud
- Click / don't click
- Success / failure

These real-life binary situations can be modeled as modified coin-flipping experiments.

3. Chi-Square Distribution

The **Chi-square distribution** is associated with the **chi-square statistic**, which measures the extent to which observed values differ from what would be expected under a **null model**. It is typically used with **counts falling into categories**. BAD702-module-2-textbook(1)

Chi-square Statistic

The general form is:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

where:

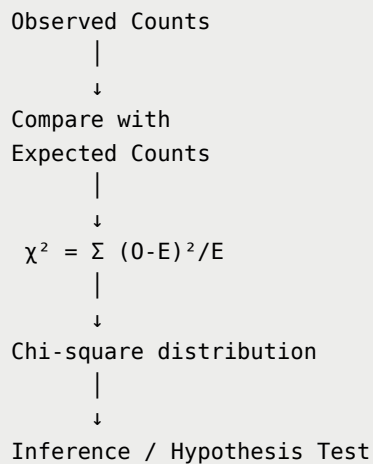
- O_i = observed value/count
- E_i = expected value/count under the null model.

Interpretation

- **Low χ^2** → observed counts are close to expected counts.
- **High χ^2** → observed counts differ substantially from expected counts.

BAD702-module-2-textbook(1)

Diagram



Applications

The Chi-square distribution is used for:

1. **Goodness-of-fit tests** — determining whether observed values fit a specified distribution.
2. **Tests of independence** — determining whether two categorical variables are independent.
3. Comparing multiple treatments when the observations are **counts**.
4. **Hypothesis testing** involving categorical data.

For example, it can test whether **gender and promotion status** are independent using the counts in a contingency table.

Different Chi-square distributions correspond to different **degrees of freedom**.

4. F-Distribution

The **F-distribution** is used primarily with experiments and **linear models involving measured continuous data**.

The **F-statistic** compares the variability **among group means** with the variability **within groups**.

Formula

In the context of ANOVA:

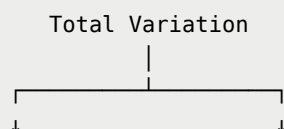
$$F = \frac{\text{Variability among groups}}{\text{Variability within groups}}$$

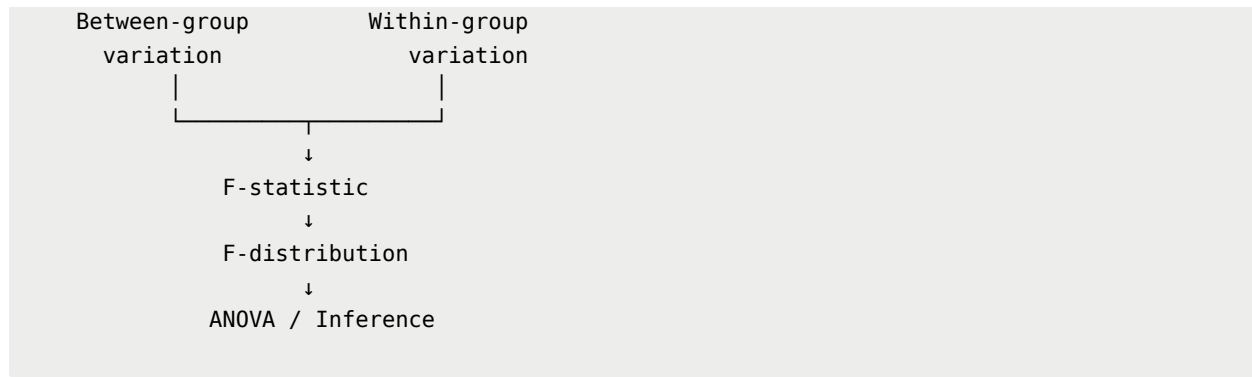
More specifically:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

where *MS* represents **mean square**.

Diagram





Interpretation

A **large F-statistic** indicates that the differences among group means are large relative to the variation within the groups.

This provides evidence that the group means may differ more than would be expected from ordinary random variation.

Applications

The F-distribution is used in:

- **Analysis of Variance (ANOVA).**
- Experiments comparing **multiple treatments or groups.**
- **Linear regression**, where the F-statistic compares the variation accounted for by the regression model with the overall variation in the data.

For example, different fertilizers can be applied to different blocks of a field, and ANOVA can determine whether the differences among group means are greater than expected from random variation.

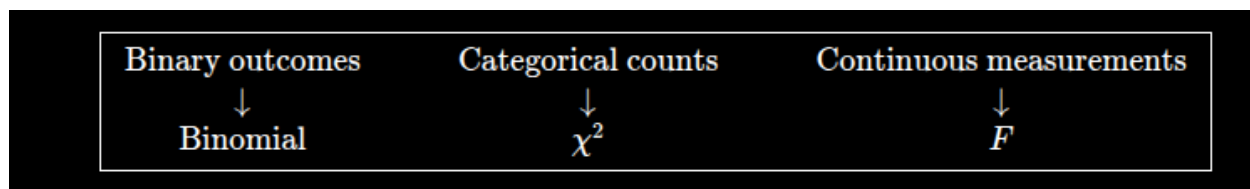
5. Comparison of the Distributions

Feature	Binomial	Chi-square	F-distribution
Type	Discrete	Continuous	Continuous
Main purpose	Number of successes	Departure from expected counts	Comparison of variances/variation

Feature	Binomial	Chi-square	F-distribution
Data	Binary outcomes	Categorical counts	Measured continuous data
Important parameter	n, pn, p	Degrees of freedom	Degrees of freedom
Main applications	Success/failure, yes/no	Goodness-of-fit, independence	ANOVA, regression
Example	Number of sales from clicks	Gender vs promotion	Comparing treatment means

6. Importance in Statistical Modeling

These distributions serve different purposes:



- **Binomial** models the number of successes in repeated binary trials.
- **Chi-square** measures how observed categorical counts depart from expected counts under a null model.
- **F-distribution** is used to compare variation among groups with variation within groups, particularly in ANOVA and regression.